INCH-POUND

MIL-PRF-81757/7D
12 September 2024
SUPERSEDING
MIL-B-81757/7C
5 October 1984

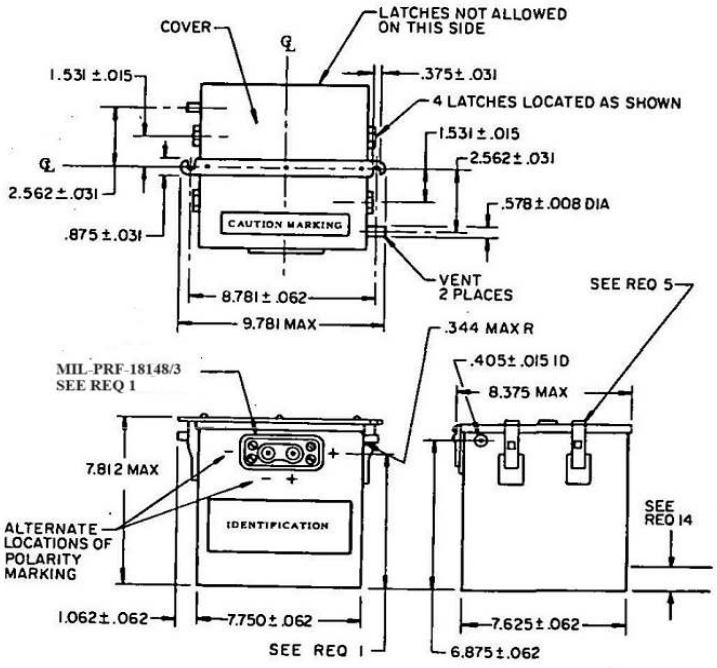
PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM, 24 VOLTS, 10 AMPERE-HOUR (BB-432A/A)

Inactive for new design after 30 July 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the battery assembly described herein shall consist of this document and the latest issue of specification MIL-PRF-81757.



NOTE: For part numbers M81757/7-2 and M81757/7-3, see requirements 15 and 16.

FIGURE 1. Battery assembly.

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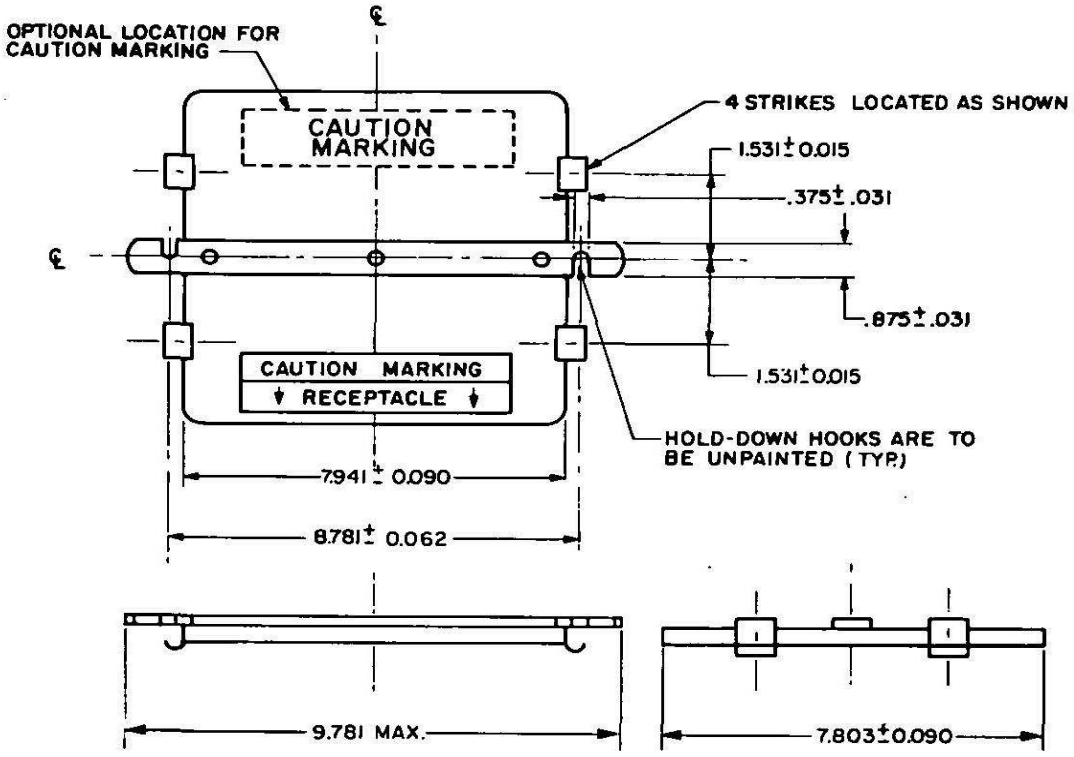
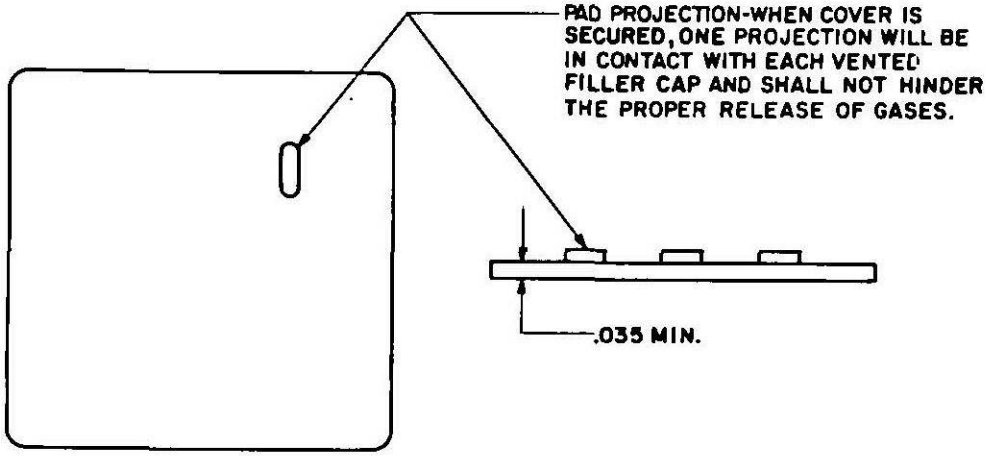


FIGURE 2. Cover.



MATERIAL: ELASTOMER (HOLD-DOWN PAD)

FIGURE 3. Cover hold-down pad.

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TABLE I. Connector strap arrangement.

Reference letter	Strap part number	Description	Quantity
A	M81757/3-1	Connector, Strap	7
B	M81757/3-2	Connector, Strap	7
C	M81757/3-3	Connector, Strap	1
D	M81757/3-4	Connector, Strap	1
E	M81757/4-1	Connector, 2 Tab	3
F	M81757/3-10	Connector, Strap	1

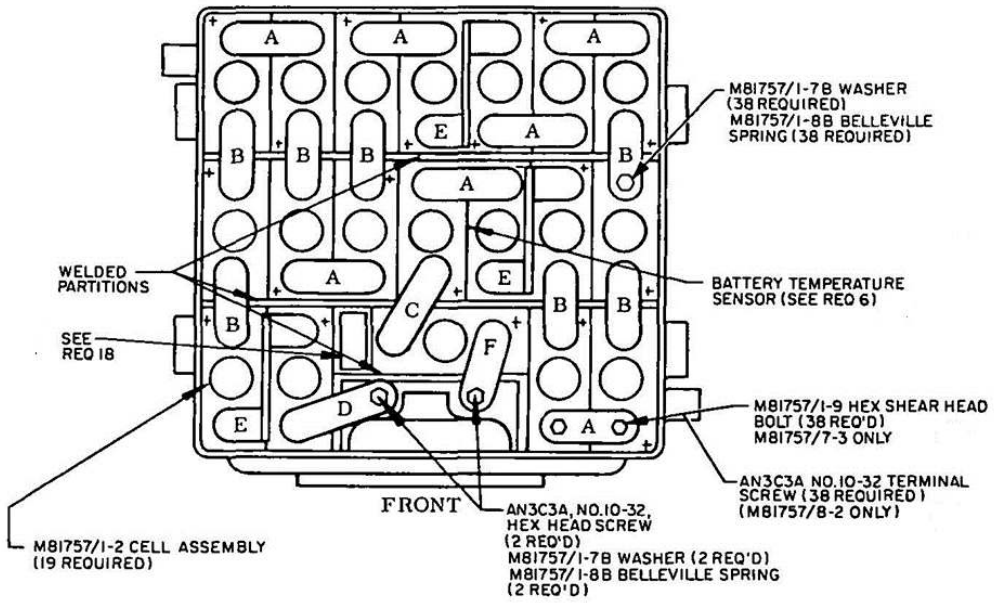


FIGURE 4. Cell arrangement.

REQUIREMENTS:

1. The receptacle terminal block shall be 0.866 ±0.015 from the centerline of the contact pins to the top surface of the terminal block. The receptacle will be so located that the top of the terminal block and the cell terminal contact surfaces are in the same plane. Receptacle terminal block shall be screw size 10-32. Receptacle shall conform to MIL-PRF-18148/3, part number MS3509-28.

2. All dimensions are prior to epoxy coating except: overall length -10.060 inches max., overall width - 8.375 inches max and overall height - 7.812 inches max.

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3. Method of mounting latches and strikes shall be optional. They shall conform to Corbin Cabinet Lock Division, American Hardware, Item Number 15834, or Nielson Latch Number C-833314-2-SS-302 or equivalent.

4. Spacers, if used, shall be polyethylene sheet in accordance with ASTM-D4066 Type I and Type III, minimum 0.015 inch thick, color natural.

5. Latches and strikes shall not be epoxy coated.

6. A sensor (thermocouple or thermistor), when required, shall be located as shown on figure 4 for the purpose of monitoring battery temperature during testing to this specification. The sensor shall be positioned in the approximate center of the cell container's widest outer surface (equidistant from the top and bottom and from side to side).

7. The battery voltage shall be 24 volts (nominal).

8. The battery weight shall be 34 pounds maximum when filled.

9. The battery shall contain 19 cells.

10. The rated capacity at 1C rate shall be 10 A.H./1·Hr./80°F/18.0V.

11. Length and width of cover pad and spacer shall conform to the inside dimensions of the applicable cover.

12. The length, width, and thickness of cover pad projections shall provide sufficient pressure on the vented filler caps to hold down the battery cells during vibration.

13. The quantity and location of cover projection pads shall be determined by the vented filler caps.

14. The base dimensions of the container shall remain within tolerance for a distance of 1.8 inches up from the bottom.

15. The M81757/7-2 battery is not for Navy use.

16. The battery identification on the M81757/7-3 battery shall indicate that P/N M81757/7-2 is superseded by P/N M81757/7-3.

17. The M81757/7-3 battery shall incorporate M81757/1-9 bolts to secure the cell interconnecting strap thereby preventing cell removal.

18. The "CAUTION" label on the M81757/7-3 battery shall contain the following information: "The cells in this battery are not to be removed."

19. The cell vented filler cap shall be in accordance with MIL-PRF-81757/2.

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20. A removable spacer 1.07 ±.03 x 0.84 ±.03 inches x 6.125 ±.03 inches shall be installed as indicated on figure 4. The spacer shall be of solid inert material, nonconductive and impervious to the electrolyte. The spacer shall be easily removable with normal shop tools.

NOTES:

l. All dimensions are in inches unless otherwise specified.

2. Supersession data: This specification sheet supersedes MS18045, MS24496, MS90321, and MIL-PRF-81757-part numbers as follows, when a product has been qualified for each MIL-PRF-81757-part number respectively.

Part Number	Replaces Part Number
M81757/7-2	MS24496-1, MS24496-2
M81757/7-3	MS18045-44, MS18045-48, M81757/7-2 and MS90321-66W

Part Number	Superseded by
MS18045-44	M81757/7-3
MS18045-48	M81757/7-3
MS24496-l	M81757/7-2
MS24496-2	MSl757/7-2
MS90321-66W	M81757/7-3
MSl757/7-2	M81757/7-3 (for Navy use only)

3. CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

4. Referenced documents: In addition to MIL-PRF-81757, this document references the following:

ASTM-D4066
MIL-PRF-18148/3
MIL-PRF-81757/2

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force – 20
DLA - CC

Preparing Activity:
Navy - AS

(Project 6140-2024-008)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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